David Stewart

Atlanta, GA | david.y.stewart1@gmail.com | linkedin.com/in/david-y-stewart | dstew2816.github.io | **U.S. Citizen**

Education

Georgia Institute of Technology - B.S. in Industrial Engineering | GPA: 3.6 Graduation: May 2028

Stamps President's Scholar - Full-ride scholarship awarded to the top 1% of applicants
Concentration: Analytics & Data Science | **Coursework**: Database Systems, Engineering Optimization, Simulation Analysis & Design, Quality Improvement Methods, Regression & Forecasting

Experience

Automation Program Management Intern May 2026 - August 2026
TSMC (Taiwan Semiconductor Manufacturing Co.) | Intelligent Manufacturing Engineer Intern *Phoenix, AZ*

- Pinpointed undetected automation failures and independently proposed, scoped, and led a technical program auditing **130** automated procedures across **2,200+** tools at the $165B Arizona fab, uncovering a **4–6%** silent failure rate.
- Translated inconsistent procedure templates into detection specs with owners across departments and Taiwan-based teams in Mandarin, writing SQL and Python over **1,000,000+** daily logs with on-call alerts for critical misses.
- Identified ∼**600 hours/month** of avoidable manual work and an estimated **six figures** in scrap risk at ∼$25K per wafer, prioritized fixes that protect wafer flow, and built a reusable audit template for new automations.
- Shipped a platform-wide **time-saved-per-run KPI** quantifying the hours each automation saves and 4 Power BI dashboards, now in **active use**, after **winning approval** by presenting to Fab 21's head of manufacturing.

Product Management Fellow September 2026 - Present
Microsoft Student Research *Remote*

- Prototyping an AI-native incident management portal for a Microsoft product team, providing enterprise customers live escalation SLA tracking, engineer messaging, configuration change alerts, and service value reporting.
- Developing the prototype as a React web app deployed on Azure Static Web Apps through a shared GitHub repository, with role-based customer and engineer views refined weekly with the Microsoft product lead.
- Benchmarking a competitor platform across **6** evaluation tracks, from on-call alerting to AI investigation agents and pricing, separating verified capability from vendor marketing with sourced evidence.

Director of Events August 2026 - Present
Product @ GT (Partners: Google, Microsoft, Home Depot, UPS)

Finance Director, Product Consultant September 2024 - Present
180 Degrees Consulting (Client: Pro Bono Partnership of Atlanta)

- Drove a **30%** increase in monthly active users and **20%** in client registration year over year by auditing engagement data and directing a content management system rebuild.

Strategy and Data Analyst September 2024 - Present
Consult Your Community

- Identified **$1M** in idle inventory for an educational distributor by digitizing legacy paper records into a unified inventory database, then delivered a B2B redirection plan.

Projects

More projects at dstew2816.github.io

OpsAudit | *Python, pandas, scikit-learn, SQL, Anthropic API* | dstew2816.github.io/ops-audit Aug 2026 - Oct 2026

- Surfaced a **26.1%** miss rate invisible to the source platform by reconstructing expected automation actions from synthetic factory history and reconciling them against execution logs.
- Built an impact model sizing **24** misses at **17.6** labor hours and **$3,618** at risk over two weeks, ranking fixes by cost to prioritize the highest-value gaps.
- Trained machine learning classifiers in scikit-learn that identify which of **33** checks a procedure monitors; a random forest scored **89%** on held-out data versus **66%** for hand-written rules.
- Implemented statistical process control with limits fixed from a healthy baseline, and stress-tested the approach on **1,567** real fab runs from UCI SECOM with every modeling choice locked in advance.

Leadership and Skills

Leadership: VP of Philanthropy, Delta Chi | CSA Finance Board | Germination Project Fellow
Technical: SQL / MySQL, Python, machine learning with scikit-learn, Power BI, Tableau, Excel, GitHub, Jira, LLM APIs
Program: Cross-functional program coordination, stakeholder management, requirements gathering, risk prioritization, root cause analysis, process mapping, process automation, statistical process control, process capability, KPI definition
Languages and Additional: Fluent in English and Mandarin Chinese | 6 years of commercial and film acting experience